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CNG AI INNOVATION FELLOWSHIP · 2026–2027

Designing Human-Centered Learning in AI-Rich Classrooms

Orchestrating Agency, Judgment, and Interpretable Evidence of Learning

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Fellowship Proposal · Academic Year 2026–2027

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*Submitted in response to the CNG AI Innovation Fellowship Call, 2026–2027
Colegio Nueva Granada AI Core Team · Proposed by Mr. Andrew Cajina*

Fellowship Thesis

In an AI-rich world, the central challenge of assessment is no longer evaluating what students produce, but interpreting evidence of how they think. At the center of the fellowship is one assessment question: if AI can produce the answer, how do we assess the reasoning, interpretation, revision, and judgment behind it? The primary question: if AI can produce the answer, how do we assess the reasoning, interpretation, revision, and judgment behind it?

Table of Contents

- 1. Executive Summary
- 2. The Fellowship at a Glance: What Happens
- 3. Why This Fellowship Exists: What Became Visible.....
- 4. Problem of Practice.....
 - 4.1 The Assessment Validity Crisis
 - 4.2 Why “AI-Rich Classrooms”: The Information Ecosystem
 - 4.3 AI as a Sociotechnical System
 - 4.4 The Invisibility of Student Intellectual Labour
 - 4.5 The SAIL Coherence Problem
 - 4.6 The Teacher Confidence Gap and the February Problem
 - 4.7 The Equity Dimension
- 5. Theoretical Framework.....
 - 5.1 Fellowship Logic Chain
 - 5.2 HCAP — The Orienting Philosophy
 - 5.3 Core Constructs
 - 5.4 Teacher Knowledge Framework
 - 5.5 Understanding the Technology: GenAI Foundations
 - 5.6 Mr. Cajina’s Role: Year 1 Facilitation
- 6. Research Base.....
 - 6.1 Assessment Design and Validity
 - 6.2 Mind, Brain, and Education: Science of Learning Foundation
 - 6.3 Validity Argument and Measurement
 - 6.4 Departmental PLCs as the Long-Term Unit
 - 6.5 Disciplinary Sciences of Teaching and I-CK Development
 - 6.6 Extended TPACK Framework
 - 6.7 CNG Pilot Evidence
- 7. Fellowship Design.....
 - 7.1 Structure
 - 7.2 Year 1 Cohort — Six Teachers, Three Mixed Pairs
 - 7.3 Two-Cycle Structure
 - 7.4 Video-Stimulated Recall — The Central Reflection Mechanism
 - 7.5 The Augmented Human Thinking Routine
 - 7.6 The Fellowship as a Learning Arc — Key Moments and Milestones
 - 7.7 Communication Check-In Schedule
 - 7.8 Decision Rules for Visible Reasoning
 - 7.9 The Right to Say No — Non-AI Pathways, Student Consent, and Agency
 - 7.10 Multiple Ways of Knowing
 - 7.11 Teacher-Friendly Language

8. CNG Tool Ecosystem.....

8.1 Technology Position Statement

8.2 Backwards Design as the Iteration Logic

9. Cognitive Agency Knowledge Hub

10. Departmental PLC and Curriculum Integration.....

10.1 Why Departments Are the Right Unit

10.2 Department Cognitive Agency PLC Protocol

10.3 Curriculum Review Integration

10.4 Emerging Shared Language — Co-Constructed Through Inquiry

11. Deliverables

11.1 Per Teacher

11.2 Fellowship Cohort Products

11.3 Institutional Products

12. Evaluation and Research Design

12.1 Primary Research Question

12.2 Research Design Matrix

12.3 Validity Strategy — Triangulated Evidence

12.4 Scope of Year 1

13. Sustainability and Institutional Rollout

13.1 The Long-Term Vision

13.2 Year 2 Architecture: From Individual to Collective TPACK

13.3 Scaling Model

13.4 Change Management

14. Direct Alignment with CNG School Priorities.....

15. References.....

Appendices A–F

“If AI can do the work and produce the product, what should remain human?”

This question drives the fellowship. It shifts the conversation from “how do we stop AI” to “what do we value.”

1. Executive Summary

The provocation. If AI can do the work and produce the product, what should remain human? This question shifts the conversation from “how do we stop AI” to “what do we value.” It applies in every subject, at every level.

The problem of practice. Teachers no longer know what evidence of learning is worth trusting when AI can generate increasingly sophisticated products. A polished essay, a solved problem set, a completed lab report: these used to function as proxies for understanding. They no longer reliably do. The deeper challenge is not how to stop AI — it is how to determine what evidence of learning actually matters and how to design assessments that generate it.

Why AI creates urgency, not the underlying problem. Assessment researchers have warned for decades that products are weak proxies for learning. Messick (1989) called it a construct validity problem. Boud and Falchikov (2007) called it the failure of sustainable assessment. Dehaene (2020) showed that products often bypass the neurological conditions for encoding. The problem predates AI. What AI did was make it impossible to keep ignoring — because the proxies have become unreliable for almost every task we used to trust.

Evidence from classroom pilots. Pilot work across Mr. Cajina’s mathematics classrooms (Systems Under Pressure, Statistical Conspiracy, the $e/\ln$ extension dashboard, the Bogotá water-crisis PBL unit) revealed a consistent pattern: when assessments were redesigned to surface reasoning, interpretation, and revision, what teachers could see about student learning changed substantially. When they were not redesigned, AI-mediated work obscured what students actually understood. This fellowship scales that approach.

The fellowship response. Six High School teachers, three genuinely mixed pairs, two cycles of assessment redesign across one academic year. Each teacher selects an assessment likely vulnerable to AI completion, audits it, redesigns it, teaches it, collects evidence of student reasoning, and reflects on what changed. They do this twice — once with structured support, once independently. Six 1-hour lunch PLCs structure the cohort’s collective learning. Two Video-Stimulated Recall sessions examine the gap between intended and enacted design. The capacity being developed: teacher judgment about what evidence of learning is worth trusting and how to design assessments that generate it.

The Innovation Architecture

VISTA (Visibility Into Student Thinking and Agency) — the framework. VISTA Audit — the assessment analysis protocol. The Augmented Human Thinking Routine — the design and reflection tool that builds AI-TPACK through structured experience. Video-Stimulated Recall — the reflection mechanism. Department PLC — the Year 2 sustainability mechanism. Each plays a distinct role in the fellowship.

2. The Fellowship at a Glance: What Happens

Before the theoretical grounding and detailed design, here is what Year 1 actually looks like.

Moment	What Happens
August — Launch	Full cohort session (60 min). Cajina models VISTA live on his own assessment. Each pair identifies their assessment.
Sep–Nov — Cycle 1	Each teacher completes their initial VISTA audit. Cajina reviews before enactment. Co-planning session: Cajina on AI-TPACK, teacher on content and pedagogy. Co-teaching or co-observation where appropriate. Student reasoning artifacts collected.
November — PLC 1	Pairs share what the artifacts revealed. Each teacher names one redesign priority for Cycle 2.
February — VSR 1	Each teacher reviews a short classroom clip with Cajina. What did I think would happen here, and what actually happened? The gap between intention and enactment is the learning.
Feb–Apr — Cycle 2	Teacher redesigns and teaches independently. Cajina responds on request. VSR Session 2 is peer-facilitated by the pair partner, with Cajina present as a silent moderator.
May–June — Transfer	Each teacher brings their two-cycle arc to their own department — iteration records, VISTA artifacts, student reasoning evidence. Cajina documents what transfers.

Two Things to Keep in Mind

Teachers lead content and pedagogy throughout. Cajina contributes AI-specific design knowledge. And VISTA audits one assessment each teacher already uses — the work is refinement of existing practice, not addition to it.

3. Why This Fellowship Exists: What Became Visible

This fellowship grew directly from classroom observations at CNG. Before the fellowship existed as a concept, two AI-rich mathematics projects surfaced a pattern that demanded a more systematic response: when students were given structured opportunities to make their reasoning visible, teachers learned things about student understanding that would have remained entirely hidden through traditional assessment alone.

In a statistics investigation, students used histograms, box plots, and measures of center to intentionally construct **competing claims from the same dataset**. Rather than simply learning statistical procedures, students began asking questions that mathematics class had not previously made space for: Who collected this data? What story is being told? What story is being hidden? How can mathematically accurate representations still be misleading? One student reflected that they would *never look at graphs the same way again*. They were not consuming information. They were evaluating how information is created and used.

AI played an unexpected role. Students had access to Flint, Gemini, ChatGPT, and NotebookLM. A consistent pattern emerged: **many students chose Flint over more powerful generative systems** — and their reason was consistent: Flint pushed them to think rather than simply providing answers.

This observation cannot be explained by Flint’s features alone. What it has is a pedagogical stance: it asks students to go further, justify more, think again. Students chose that stance over efficiency. That kind of choice requires a classroom culture in which deeper thinking has been established as genuinely valuable — where the relationship between students and teacher, and between students and the work itself, has created conditions in which intellectual challenge is worth choosing. Students also shared what they were doing with their teacher and with each other. That sharing was relational: the work mattered partly because the relationships around it made it matter.

This is a finding about **trust, relationship, and the conditions under which student agency is exercised toward learning** rather than away from it. The design implications are not primarily about tool selection. They are about what a classroom must be before tool choice becomes an expression of student agency.

A second project, Systems Under Pressure, revealed that AI could extend mathematical inquiry rather than replace it. Students investigated authentic systems — energy markets, water supply networks, food systems — using sequences, exponential functions, logarithms, and polynomial behavior. AI allowed them to ask better mathematical questions: How might this system be represented? Which variables matter? Which aspects resist modeling, and why? For many students, mathematics shifted from a collection of disconnected topics to a connected web of relationships and interpretations.

Visible reasoning protocols — reasoning vlogs, annotated drafts, AI disclosure reflections, structured questioning — gave teachers access to evidence that final products could not reveal: where students changed their minds; where AI deepened understanding and where it replaced it; which misconceptions persisted; how students justified accepting or rejecting AI-generated ideas. These moments were invisible in final products but visible in the reasoning process. **The fellowship is the systematic response to these observations.** Its purpose is not to prove that AI improves learning. Its purpose is to help teachers design learning environments where the development of judgment, disciplinary understanding, and student agency remains visible even when AI can generate increasingly sophisticated products.

Note on positionality: Cajina is both the facilitator and the developer of the VISTA Protocol. This recognised condition in practitioner inquiry is managed through transparent documentation, triangulated evidence, and participant feedback at each cycle.

4. Problem of Practice

4.1 The Assessment Validity Crisis

In classrooms where students have access to AI tools, something has changed about what a finished piece of work tells a teacher. A polished essay, a solved problem set, a completed lab report: the product no longer reliably shows whether the student did the thinking. A student who outsourced the work and one who genuinely understood it can produce indistinguishable outputs. **This is not a new problem. AI made it impossible to keep ignoring.**

What we have long called ‘evidence of learning’ was often evidence of something else: task completion, genre fluency, deadline compliance. Messick (1989) identified this as a construct validity problem — we assess proxies rather than the thing itself. Boud and Falchikov (2007) showed that students who cannot evaluate their own work have not learned in any durable sense. Black and Wiliam (1998) demonstrated that formative feedback only produces learning when students use it to close their own gap in understanding — a gap AI can now close for them. Dehaene (2020) is unambiguous: active engagement, effortful retrieval, and error feedback are neurological requirements for encoding, not pedagogical preferences. AI-bypassed tasks eliminate all three. Whitman and Kelleher (2016) name the same through the lens of desirable difficulty: productive struggle is not an obstacle to learning. It is the mechanism.

Phillip Dawson (2021) synthesises it: AI does not create an assessment crisis — it reveals one that already existed. The crisis is the long-standing gap between what we assess and what actually constitutes learning. The real loss has been **dialogue, reasoning, voice, and judgment** — the processes through which learning actually happens, increasingly invisible behind finished products. The question this fellowship addresses is not how to protect assessment from AI. It is how to design assessment that generates evidence of those processes — evidence that holds up regardless of what tools students use.

The future of assessment is not protecting products from AI. It is making reasoning, interpretation, revision, and judgment visible *despite* AI — and because of what AI reveals about what we were never assessing well enough in the first place.

This fellowship develops teacher capacity to do exactly that. Not capacity to use AI tools, manage AI policy, or detect AI use — but the capacity to look at any assessment and ask: which learning processes does this task actually demand? Are those processes currently visible? Can I collect evidence of them when students have access to AI? Teachers who can answer those questions can design for learning that holds up regardless of what tools students use. That judgment is what this fellowship builds.

4.2 Why “AI-Rich Classrooms”: The Information Ecosystem

The fellowship uses the term **AI-rich classrooms** intentionally. An AI-rich classroom is not simply one where teachers actively use AI tools. It is a classroom situated within a broader information ecosystem increasingly shaped by AI. Students encounter AI through search engines, recommendation systems, translation tools, writing assistants, social media feeds, generative AI platforms, and emerging AI agents. Even when AI is not explicitly part of a lesson, it often influences how information is found, organized, interpreted, and communicated.

Recent reports suggest that automated systems and AI agents now generate a substantial proportion of global web traffic — in some cases exceeding human-generated activity. The broader trend is clear: AI is becoming part of the infrastructure through which people engage with knowledge online. The question for schools is no longer whether AI is present. The question is how educators design human-centered learning experiences within an environment where AI increasingly shapes the conditions under which learning occurs.

This has implications for how we think about assessment validity. A student who submits an essay produced by AI is not violating a rule in a controlled environment — they are responding rationally to the information ecosystem they live in. **The design challenge is not to seal the classroom off from that ecosystem but to make the learning within it visible and meaningful despite it.** This is the problem the fellowship addresses.

4.3 AI as a Sociotechnical System

Generative AI is not a neutral tool. Winner (1980) established that artifacts have politics — technologies encode and redistribute power in ways users may not notice. Selwyn (2019) extended this to educational technology. Orlikowski (1992) demonstrated that technologies gain their effects through the social practices enacted around them — meaning thoughtful practice design can reshape what AI does in a classroom. In AI-rich classrooms, AI is reshaping who counts as a knower, whose work gets attributed, and what level of performance signals understanding.

4.4 The Invisibility of Student Intellectual Labour

Student **intellectual labour** — the cognitive work of sense-making, evaluating, revising, and constructing understanding — includes not only reasoning and explanation but **creation, embodiment, and tacit knowledge**: the artist’s aesthetic judgment, the mathematician’s structural intuition, the programmer’s debugging instinct. These forms of knowing are discipline-specific and often the hardest to develop. Product-only assessment systematically misses them. Visible reasoning protocols make them interpretable.

4.5 The SAIL Coherence Problem

CNG’s SAIL framework is strong policy architecture. But the same SAIL level can mean very different things across classrooms without a shared pedagogical mechanism. Perkins et al. (2024, 2025) argue that AI-use continua become educationally meaningful only when tied to intended learning outcomes and assessment design. Choosing a SAIL level is a simultaneous intersection of AI-TPACK (Can I?), Ethical TPACK (Should I?), XK (Will it work here?), and HAIC-K (How?). The fellowship develops that four-layer judgment deliberately.

4.6 The Teacher Confidence Gap and the February Problem

Meishar-Tal et al. (2025) find that teachers’ classroom AI use is predicted primarily by *perceived behavioral control* — confidence in the ability to use AI pedagogically. Design capacity is what predicts enactment.

If I am a tired teacher in February, what am I actually doing differently?

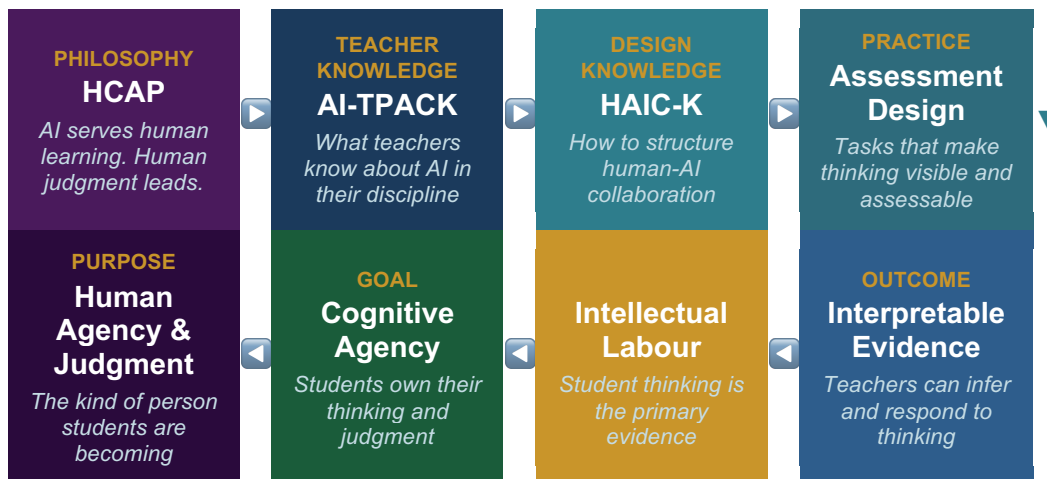
The answer the fellowship builds toward is: one question. Before designing a task — or when reviewing one already in place — ask: if AI completed this perfectly, what learning would disappear? That takes thirty seconds. It requires no new tools, no new curriculum, and no additional meeting. Over time, repeated use of this question and the Department Cognitive Agency PLC Protocol embeds new assessment judgment into existing routines. The fellowship is designed for the tired teacher in February, not only for the enthusiastic teacher in September.

4.7 The Equity Dimension

In high-resource environments like CNG, generative AI risks becoming invisible advantage. Benjamin (2019) and Birhane (2021) demonstrate that algorithmic systems used uncritically reproduce and amplify existing inequities. Holstein and Doroudi (2022) extend this to educational AI. The fellowship addresses this by making intellectual labour — in all its forms, across all modes of expression — the primary evidence of learning. Universal Design for Learning (CAST, 2018) provides the accessibility framework: designing for full participation from the outset, not retrofitting accommodations.

5. Theoretical Framework

5.1 Fellowship Logic Chain



5.2 HCAP — The Orienting Philosophy

Human-Centered AI Pedagogy (HCAP; Chiu, 2026) is the orienting philosophy: AI serves human learning rather than replacing it, teacher judgment remains primary, and student cognitive agency is non-negotiable. Every design decision in the fellowship is evaluated against the HCAP question: *does this serve human learning, or does it substitute for it?* HCAP aligns directly with CNG’s own philosophy statement: *Universal moral values such as respect, responsibility, integrity, and caring, are taught through direct instruction, modeling, and experience* (CNG Philosophy). A school that states this cannot treat AI as a values-neutral tool.

5.3 Core Constructs

Student Cognitive Agency

Student Cognitive Agency is the capacity of learners to act as primary sense-makers, evaluators, decision-makers, and architects of understanding. Students with developed cognitive agency use AI as advisory intelligence — interrogated, challenged, verified, selectively incorporated. They retain final judgment.

Measurement note: Cognitive agency is a latent construct — inferred, not directly observed. The fellowship does not claim to directly measure agency. Instead it triangulates **indicators of cognitive agency** across multiple evidence sources: student-led hypothesis testing, independent question generation, strategic AI use, justified rejection of AI suggestions, and explanation of decision-making. These indicators are interpreted collectively rather than treated as direct measures.

Student Intellectual Labour and Human Agency and Judgment

Student Intellectual Labour refers to the full range of cognitive and creative work students perform — including reasoning, **creation, embodiment, and tacit knowledge**. This work has intrinsic educational value independent of final product quality.

The deeper purpose is **Human Agency and Judgment**: the kind of person students are becoming — capable of inquiry, critique, disciplinary decision-making, and responsible AI use. This is the fellowship’s horizon, connecting directly to CNG’s vision of *leadership and service in the world of today for a better tomorrow*.

Interpretable Evidence of Learning

Artifacts that allow teachers to infer, diagnose, and respond to student thinking processes — not merely evaluate final products. Evidence takes multiple forms: verbal, written, visual, computational, **embodied, artistic, and creative**. The quality of reasoning is assessed — not the mode of expression.

Some of the most important learning is not fully visible. The fellowship does not claim interpretable evidence captures everything that matters — only that it captures more than polished product alone. This connects naturally to **standards-based grading** practices: the fellowship’s rubric dimensions describe developmental progressions informed by Modern Measurement Theory and construct-map thinking (Saubern et al.). While Year 1 will not generate sufficient data for formal Rasch modeling, the framework adopts the underlying principle of developmental continua — a natural companion to CNG’s movement toward standards-based grading.

SAIL as a TPACK Decision — Noticing What Teachers Already Know

CNG teachers already make CK, PK, and TK intersections every day. The fellowship helps teachers *notice, name, and extend* that integration into AI contexts. Mr. Cajina’s role in PLCs and VSR cycles is to function as a *mirror*: surfacing decisions teachers are already making and extending that navigation deliberately.

5.4 Teacher Knowledge Framework

The fellowship develops four AI-specific knowledge layers — building on the **CK, PK, and PCK that departments already share**:

Knowledge Layer	The Question	What It Requires
Layer 1 — AI-TPACK (Celik, 2023; Chiu, 2025)	<i>Can I integrate AI?</i>	Integrated knowledge of AI tools, disciplinary content, and pedagogy — including I-CK (how AI handles the specific knowledge structures of a discipline)
Layer 2 — Ethical TPACK (Celik, 2023; UNESCO, 2024; Chiu, 2026)	<i>Should I integrate AI?</i>	Knowledge to identify bias in AI outputs, evaluate hallucinations, protect student data privacy, preserve academic integrity, and support equitable participation
Layer 3 — Contextual Knowledge (XK) (Porras-Hernández & Salinas-Amescua, 2013)	<i>Will this work here?</i>	Understanding of school culture, student demographics, curriculum requirements, assessment pressures, available tools, and institutional policy. Departments already share CK, PK, and PCK — XK is what the fellowship makes explicit for AI contexts
Layer 4 — HAIC-K (Chiu, 2026)	<i>How should humans and AI collaborate?</i>	Human-AI Collaboration Knowledge: role allocation, co-agency, human oversight, verification, critique structures, and epistemic vigilance so students remain primary agents

These four layers operate *simultaneously*, not sequentially. Critically, **departments already share CK, PK, and PCK** — the foundational knowledge that defines a discipline and how it is taught. The fellowship does not rebuild what departments already know. It builds AI-specific knowledge on top of that shared foundation. This is precisely why the departmental PLC is the right long-term unit for sustainability.

5.5 Understanding the Technology: GenAI Foundations for Teacher Conversations

A central goal of the fellowship is for teachers to develop the knowledge needed to navigate AI with informed judgment rather than policy compliance. This requires knowing something about how generative AI actually works — not at an engineering level, but at the level that shapes pedagogical decision-making.

Generative AI systems are **probabilistic text predictors**. They predict statistically likely sequences of tokens based on patterns in training data. When a student submits a writing prompt, the system generates a statistically plausible response — one that *looks like* responses it has seen, assembled word by word based on probability distributions. Bender et al. (2021) coined the phrase **stochastic parrot** to describe this: a system that produces fluent, contextually appropriate-seeming language without any grounding in meaning, understanding, or truth. This is not a flaw to be fixed. It is the nature of how these systems work.

Teachers and students also need to understand the **black box problem**. Most frontier AI systems are opaque: their internal processes are not interpretable even to the teams that built them. When a teacher asks why an AI gave a particular answer, there is no reliable explanation. The system cannot tell you. This is not evasiveness — it is an architectural reality. This opacity is precisely why **navigating ambiguity** is a core professional competency for the AI era: teachers must make disciplinary judgment calls about AI outputs without being able to fully verify the reasoning behind them. The VISTA Protocol is a practical mechanism for exercising that judgment in assessment design.

The fellowship addresses these foundations not through a workshop but through **Cajina-facilitated conversations as teachers encounter them in practice**. When a teacher audits a task and notices that AI produces fluent analysis without genuine disciplinary reasoning, that is the moment to examine the stochastic parrot problem. When a teacher asks why AI confidently produced an incorrect claim, that is the moment to examine the black box. Understanding builds from disciplinary contact with the technology — the goal is teachers who can name both what AI can do and why its limits are genuinely irreducible.

5.6 Mr. Cajina's Role: Year 1 Facilitation

In Year 1, Mr. Cajina holds the facilitation role across three functions: **iteration feedback** (written feedback on each redesign before enactment), **epistemological tension facilitation** (surfacing the hard questions AI raises in each discipline — not to resolve them, but to make them discussable), and **ethical tension modeling** (reasoning through bias, privacy, and epistemic vigilance through disciplinary cases, not policy recitation).

From Year 2 onward, the **CNG AI Task Force** assumes the facilitation role. The fellowship produces an AI Task Force Handoff Package — ensuring the model is replicable without dependency on any single person. Year 1 participants who wish to contribute may do so; no assumption of volunteer co-facilitation is made.

6. Research Base

6.1 Assessment Design and Validity

Constructive alignment (Biggs, 2003) requires coherence among learning objectives, instructional activities, and assessment tasks. In AI-rich environments this breaks when assessments evaluate products AI can generate rather than the reasoning they were designed to elicit. Black and Wiliam (1998) established that formative assessment requires access to evidence of student thinking. Wiggins (1998) argued for authentic assessments requiring performance of actual disciplinary work. Pellegrino et al. (2001) articulated the assessment triangle — cognition, observation, interpretation. The VISTA Protocol operationalizes the observation vertex.

Dawson (2021) provides the intellectual foundation for this fellowship’s orientation. His argument: AI does not threaten assessment — it exposes how far assessment has drifted from its core purpose. When AI can produce the outputs we evaluate, we are forced to ask what we were actually assessing. His answer is that assessment’s human core consists of **dialogue, reasoning, voice, and judgment** — four dimensions this fellowship makes its design target.

The fellowship’s VSR component is grounded in documented research on the **know-do gap** — the persistent discrepancy between teachers’ planned technology integration (Intended TPACK) and their actual classroom performance (Enacted TPACK). Hanifah et al. (2026) propose a triadic framework — Intended, Enacted, and Reflected TPACK — and find that 65% of planned technology-rich activities are shortened or abandoned during live enactment. Video analysis reveals that the dual cognitive load of managing technical tools while facilitating student inquiry frequently exceeds teachers’ capacity, causing regression to safer, teacher-centered methods. VSR functions as a **metacognitive mirror**: teachers reviewing their own lesson recordings move from defensive justifications toward recognizing orchestration rigidity, building the adaptive expertise required for AI-rich classrooms. Triangulating VSR data with audit artifacts and reflections — across Intended, Enacted, and Reflected TPACK — produces the most valid depiction of how AI-TPACK develops in practice.

6.2 Mind, Brain, and Education: The Science of Learning Foundation

The fellowship’s design is grounded in three bodies of science-of-learning research that translate consistently across the neuroscience, cognitive science, and educational psychology literature: Whitman and Kelleher’s *Neuroteach: Brain Science and the Future of Education* (2016); Dehaene’s *How We Learn: Why Brains Learn Better Than Any Machine — For Now* (2020); and Carey’s *How We Learn: The Surprising Truth About When, Where, and Why It Happens* (2014). Together, these works establish a set of principles that directly shape fellowship design decisions.

From **Neuroteach** (Whitman & Kelleher, 2016): the brain encodes knowledge through active retrieval, not passive reception. Desirable difficulty — tasks that require genuine effortful processing — produces stronger memory encoding than fluent, easy performance. AI-bypassed tasks eliminate the desirable difficulty that the brain needs for learning to occur. The VISTA Protocol is, at a neuroscientific level, asking: *does this task require the effortful processing that enables memory consolidation?* The flip side is the spacing effect: revisiting a redesigned assessment across multiple iterations is more powerful than a single well-designed task.

From **How We Learn** (Dehaene, 2020): four pillars of learning are attention, active engagement, error feedback, and consolidation. Dehaene argues explicitly that passive, fluent generation — the mode AI operates in — does not produce learning. Only the student who makes the errors, receives the feedback, and actively corrects their understanding is learning. This is not a metaphor: it is a neural mechanism. AI that completes tasks for students removes all four pillars simultaneously. The fellowship’s redesign pathways — reasoning vlogs, revision narratives, AI rejection logs — are designed to restore attention, active engagement, and error feedback to tasks where AI would otherwise eliminate them.

A direct connection: Dehaene’s discussion of the **consciousness of error** maps precisely onto the fellowship’s emphasis on visible reasoning. When a student documents a moment where they rejected an AI suggestion, they are engaging the very neural mechanism — prediction error — that Dehaene identifies as central to learning. The AI rejection log is not just a transparency protocol. It is a science-of-learning intervention.

The implications for AI-era assessment design are direct: if a task produces no attention, no active engagement, no error feedback, and no consolidation, it produces no learning — regardless of how polished the output looks. The VISTA Protocol’s central question is, at this level, asking whether a task still produces the conditions under which brains learn.

The fellowship is explicitly grounded in the **Science of Learning (SoL)** literature — specifically Dehaene’s *How We Learn* (2020) and Whitman and Kelleher’s *Neuroteach* (2016), which translates Mind, Brain, and Education (MBE) research into classroom-applicable principles. These sources are not background context. They are the *design rationale* for every protocol, PLC structure, and iteration cycle in the fellowship.

Whitman and Kelleher (2016) distill the MBE research into twelve principles that the fellowship operationalizes directly. The table below maps each principle to its fellowship mechanism. Critically, **TPACK is the theoretical bridge** between these principles and classroom practice: understanding how the brain learns (MBE) tells us what good learning looks like; understanding how to integrate technology purposefully into that learning (TPACK) tells us how to design the assessment and AI-use structures that support it. When the fellowship develops teachers' AI-TPACK, Ethical TPACK, XK, and HAIC-K, it is building the TPACK infrastructure that makes MBE principles actionable in AI-rich classrooms. TPACK is the foundation of that research — the bridge between learning science and instructional design.

MBE Principle (Whitman & Kelleher, 2016)	Fellowship Mechanism (TPACK Bridge)
<i>1. Prior knowledge matters: new learning builds on what learners already know</i>	The fellowship begins with the VISTA Protocol — which requires teachers to surface their existing disciplinary knowledge (CK and PCK) as the foundation for I-CK development. TPACK builds on prior content knowledge, not around it.
<i>2. Learning requires attention: focused attention is prerequisite to encoding</i>	PLC sessions are 30 minutes by design — within the attention bandwidth that produces encoding. VSR cycles focus on one specific hinge point. No session tries to cover everything.
<i>3. Learning is enhanced by emotion: moderate challenge with safety produces optimal engagement</i>	The fellowship design creates psychological safety through 1:1 written feedback before public enactment. Teachers encounter challenge (audit, redesign) in low-stakes async contexts before high-stakes classroom enactment.
<i>4. Spaced practice: distributed practice over time outperforms massed learning</i>	The year-long iteration arc — multiple design-enact-revise cycles distributed across eleven months — directly operationalizes spaced practice. Professional learning is not a workshop. It is a sustained cycle.
<i>5. Interleaving: mixing topics or problem types improves long-term retention</i>	Cross-disciplinary PLC discussions deliberately interleave perspectives from mathematics, science, humanities, and arts. Seeing how the VISTA Protocol works in another discipline strengthens understanding in one's own.
<i>6. Retrieval practice: testing oneself on material strengthens memory more than re-reading</i>	VSR cycles require teachers to retrieve and articulate their own instructional decisions from video evidence — a form of self-testing that produces deeper encoding than reflective journaling alone.

<i>7. Metacognition: awareness and regulation of one's own thinking is teachable</i>	The Metacognition rubric dimension, the Reasoning Vlog protocol, and the Revision Narrative protocol all target metacognitive development explicitly. Teachers develop metacognitive habits; students develop them through visible reasoning protocols.
<i>8. Feedback: specific, timely feedback accelerates learning</i>	Cajina provides written feedback on every redesign before enactment — not after. Feedback is specific (design quality, SAIL coherence, intellectual labour visibility) rather than general. The TPACK framework specifies what to give feedback on.
<i>9. Active engagement: passive reception does not produce durable learning</i>	Every fellowship touchpoint requires active production: audit, redesign, artifact review, VSR analysis, showcase presentation. There are no lecture sessions. The PLC Protocol culminates in one specific design change, not a discussion conclusion.
<i>10. Sleep and consolidation: learning consolidates during rest; distributed sessions outperform marathon sessions</i>	The fellowship's eleven-month distributed structure directly reflects consolidation research. Monthly touchpoints allow consolidation between active design cycles. No marathon PD days.
<i>11. Stress and learning: chronic stress impairs learning; moderate challenge supports it</i>	The Fellowship is designed for the tired teacher in January. Async feedback and design questions (the VISTA (Visibility Into Student Thinking and Agency) Protocol) are explicit workload design decisions grounded in stress-learning research.
<i>12. The learning environment matters: physical and social context shapes what is learned and retained</i>	The Department Cognitive Agency PLC Protocol is designed for the existing departmental meeting room, existing meeting time, and existing PLC culture — not a new environment. The Knowledge Hub is accessible from wherever teachers already work.

Dehaene (2020) contributes four **pillars of learning** that reinforce this design: **attention** (the fellowship structures focused, bounded sessions), **active engagement** (every touchpoint requires production), **error feedback** (iteration cycles are designed around encountering and revising based on what student artifacts reveal), and **consolidation** (the distributed eleven-month arc). When teachers develop AI-TPACK and HAIC-K through this fellowship, they are not just learning new knowledge — they are learning in the way the brain learns best: through distributed, active, feedback-rich, metacognitive practice. The PD design *enacts* the same principles it asks teachers to apply.

6.3 Validity Argument and Measurement

This fellowship adopts a **validity argument approach** (Messick, 1989; Kane, 2006): claims about learning are supported through the convergence of multiple evidence sources, not any single artifact. The strength of an interpretation increases when evidence converges across assessment design, interaction traces, student artifacts, teacher observations, and reflective accounts.

The fellowship is informed by Modern Measurement Theory (MMT) and construct-map thinking (Saubern et al.), emphasizing developmental progressions rather than binary proficiency judgments. While Year 1 will not generate sufficient data for formal Rasch modeling, the framework adopts the underlying principle of developmental continua.

A key methodological distinction runs through all measurement in this fellowship: **confidence is not competence**. Self-report surveys measure what teachers believe they can do; the Technology Integration Assessment Instrument (TIAI) measures what they actually design. The TIAI scores the ‘fit’ and alignment of technology, pedagogy, and content in observed lesson plans and assessment designs, producing a pre/post comparison that shows how AI-TPACK manifests in teachers’ actual instructional decisions across iterations. The VISTA audit artifact provides the primary evidence source for this comparison, while VSR transcripts provide the Enacted and Reflected TPACK dimensions.

On verbal fluency: The fellowship mitigates verbal fluency bias through UDL safeguards and action-based evidence. Interaction traces — revision narratives, prompt logs, creation artifacts, embodied demonstrations — demonstrate the human moves in a project regardless of verbal fluency. The score is assigned to the demonstrated decision-making arc, not the polish of the explanation.

6.4 Departmental PLCs as the Long-Term Unit

Fullan (2016) demonstrates that deep learning requires embedding new practices in collaborative professional structures. DuFour and Eaker (1998) establish that PLCs built around shared inquiry into student learning are the most powerful professional development model available. Coburn (2003) argues that sustainable reform requires *deep* change — shifting underlying beliefs and practices — most reliably produced through communities of practice. Bryk et al. (2015) demonstrate that reliable change builds on existing organizational structures. Hargreaves and Fullan (2012) identify *professional capital* — the collective knowledge, skill, and decisiveness of a teaching community — as the key variable in educational improvement.

Departments are the right unit because they already share **CK, PK, and PCK** — the foundational domains that make AI-specific knowledge meaningful. A science department that shares a commitment to evidence-based reasoning already has the PCK that makes Ethical TPACK concrete. The fellowship does not build new PLCs. It enriches existing ones with the AI-specific knowledge layers their shared CK and PK need in order to navigate AI-rich classrooms.

Research on AI-era professional development adds a further argument: AI-TPACK development is *inherently social*. Studies propose the concept of **Collective TPACK** — the shared, distributed AI-pedagogical knowledge that emerges when teachers work together on real disciplinary problems, rather than developing AI knowledge in isolation. Generic training fails because AI’s impact is highly dependent on subject matter: STEM faculty often develop different AI competencies than humanities faculty, and one-size-fits-all workshops cannot provide the discipline-sensitive support that departments naturally offer. PLCs allow teachers to collaboratively create shared norms for specific tools, ensuring AI use is aligned with departmental standards rather than individually improvised.

A specific risk the departmental structure addresses is what researchers call the **XK filter**: institutional pressures like exam pacing, curriculum requirements, and assessment timelines suppress AI innovation even in willing, capable teachers. A classroom teacher working alone cannot negotiate those pressures. A department leader in the room can. This is why the fellowship pairs department leadership with teachers from the outset — not only because department leadership can spread the protocol later, but because their positional authority creates the conditions under which genuine redesign is possible at all.

Finally, research consistently shows that teachers who engage in job-embedded professional development report higher digital self-efficacy and are more likely to perceive technology as a resource rather than a threat. The fellowship’s design — working with real assessments, real students, and specific written feedback across a sustained year-long cycle — is job-embedded by design. The PLC is not a separate learning activity. It is the site where **collective sensemaking** happens: where teachers together negotiate what AI means for their discipline, redefine their roles, and build the shared professional language that makes change durable.

6.5 Disciplinary Sciences of Teaching and I-CK Development

Every discipline has its own research-validated knowledge base about how students learn that subject. These are not generic pedagogical principles — they are discipline-specific sciences of teaching, accumulated through decades of field-specific research:

Discipline	Disciplinary Science of Teaching Foundation
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Mathematics Education	NCTM Standards; mathematics cognition research (Dehaene, 2011); productive struggle and mathematical mindsets (Boaler, 2016); algebraic reasoning and structural understanding; the Van Hiele levels of geometric thinking; ZDM Mathematics Education literature
Science Education	Next Generation Science Standards (NGSS) practices; model-based learning and model revision; Claim-Evidence-Reasoning frameworks; misconceptions research; conceptual change theory (Posner et al., 1982)
Humanities and Social Studies	Disciplinary literacy and historical thinking (Wineburg, 2001); sourcing, contextualization, and corroboration as epistemic practices; contested interpretation and evidence standards; reading like a historian
Arts Education	Studio Thinking Framework (Hetland et al., 2007); the eight Studio Habits of Mind; craft knowledge and embodied expertise; artistic decision-making as disciplinary reasoning
Language Arts and ELL	Genre-based writing; register and pragmatic competence; translanguaging (García & Wei, 2014); academic language development; disciplinary literacy across content areas

This matters for the fellowship in a specific way: **I-CK is not about knowing what AI does to generic content. It is about knowing what AI does to your discipline’s specific research-validated knowledge structures.** A mathematics teacher developing I-CK needs to recognize when AI produces a plausible-looking proof that violates the structural reasoning NCTM research identifies as foundational. A science teacher developing I-CK needs to recognize when AI generates an explanation that bypasses the model-revision process that NGSS identifies as the core of scientific practice.

This is also why the **departmental PLC is the right long-term unit**: departments already share their disciplinary science of teaching. A mathematics department that reads ZDM together and discusses misconceptions research already has the shared disciplinary knowledge base that makes I-CK development meaningful. The fellowship does not ask teachers to develop I-CK in isolation. It asks departments to apply the disciplinary knowledge they already share to a new question: *what does AI do to the knowledge structures our discipline’s research says students need to develop?*

The Cognitive Agency Knowledge Hub (NotebookLM) is therefore not only a repository of fellowship artifacts. It is a space where teachers can continuously engage with their discipline’s research literature and interrogate it against AI behavior — the ongoing I-CK development that makes the fellowship’s effects durable rather than one-off.

6.6 Extended TPACK Framework

From TPACK (Mishra & Koehler, 2006) through AI-TPACK (Celik, 2023), I-TPACK (Chiu, 2025), and the HCAP framework with HAIC-K (Chiu, 2026): in AI-rich classrooms, AI produces content-like outputs that may or may not reflect genuine disciplinary knowledge structures. **I-CK** — the knowledge of how AI handles a discipline’s epistemic standards — is what allows teachers to design valid assessments. Relational TPACK (Delanoy-informed) recognizes that teacher knowledge includes relational authority and dialogic feedback. Ethical TPACK positions ethical reasoning as an integrated pedagogical judgment, not a compliance obligation.

6.7 CNG Pilot Evidence

Systems Under Pressure — Algebra 2

Students investigated complex systems using sequences, exponentials, logarithms, and polynomial models. Without reasoning artifacts, students who used AI superficially were indistinguishable from those who used it deeply. With them, the difference was immediate and actionable for feedback. Creation artifacts — students’ own models and explanations of why their assumptions changed — revealed understanding that verbal summaries never could.

Statistical Conspiracy Project — Accelerated Geometry

Students investigated how the same dataset supports competing narratives. The most significant learning — students explaining why they rejected an AI-generated causal interpretation, drawing on distributional reasoning and sampling limitations — was completely invisible without reasoning artifacts. That moment of rejection is the template for interpretable evidence of intellectual labour.

6.8 Iteration and Improvement: How Teachers Actually Change Practice

This fellowship is built on a specific claim from the research on teacher learning: practice does not improve through exposure to ideas alone. It improves through repeated cycles of design, enactment, evidence collection, reflection, and revision.

Bryk, Gomez, Grunow, and LeMahieu (2015) on improvement science: change in complex educational systems requires structured, iterative, evidence-based cycles — not single interventions. Their PDSA (Plan-Do-Study-Act) framework underwrites the two-cycle design here. Each cycle in the fellowship is a complete PDSA loop: plan the redesign, do the enactment, study the artifacts, act on the revision.

Timperley (2011) on the Teacher Inquiry and Knowledge Building Cycle: sustained changes in teacher practice come from inquiry into specific student evidence, not from general professional development. The fellowship's VSR sessions and PLC artifact reviews are structured Timperley cycles — teachers examine what students actually did and use that evidence to revise their next attempt.

Design-Based Research (Brown, 1992; Collins, 1992) provides the methodological frame for treating the classroom as the unit of design refinement. The fellowship is not implementing a finished protocol; it is iteratively refining VISTA, the Augmented Human Thinking Routine, and the assessment redesigns through enactment in real classrooms. Each pair contributes to the refinement, and the cross-cycle evidence informs the Year 2 institutional version.

Hattie (2009, 2012) on feedback and revision: the largest learning effects come from feedback that students and teachers actually use to close a gap in their own work. The fellowship's structure operationalises this for teacher learning: the VSR gap analysis is feedback the teacher cannot ignore, because they are watching themselves teach.

The synthesis: two cycles, six PLCs, two VSR sessions, and one written-feedback pass are not arbitrary scheduling decisions. They are the minimum architecture the research on teacher improvement identifies as necessary for practice to actually change. Anything less is exposure to ideas, not capacity-building.

7. Fellowship Design

7.1 Structure

The fellowship is a year-long iterative design cycle. Each teacher completes two to three full **design** → **enact** → **evidence** → **feedback** → **revise** arcs. The total async commitment outside scheduled touchpoints is approximately 4–5 hours across eleven months — designed for the realistic demands of teaching, not only for ideal conditions.

7.2 Year 1 Cohort — Six Teachers, Three Mixed Pairs

The Year 1 cohort consists of six secondary teachers in three mixed-experience pairs. Pairs are composed the way departments actually are: teachers at different stages of AI integration, different comfort levels, different disciplinary contexts, working together. No pair is grouped by similarity. This is deliberate — the fellowship tools must work in the conditions they will actually be deployed into, which are always mixed.

Pair	Composition and Rationale
Pair A	Two teachers from different disciplines at different stages of AI integration. Mixed experience, mixed comfort. The pair dynamic surfaces disciplinary differences in how VISTA applies across subjects.
Pair B	Two teachers where one is further along with AI integration than the other. The more experienced teacher articulates tacit knowledge; the other asks the questions that make it explicit. Mutual accountability without hierarchy.
Pair C	Two teachers with genuinely different orientations toward AI in teaching — one more confident, one more uncertain. If the fellowship tools work for this pair, they work in any department.

Pairs are selected to mirror department reality. In Year 2, when teachers bring this work to their departments, they will encounter exactly this range of orientations. Year 1 pairs test the tools in conditions that resemble the scale-up environment.

Additional selection criteria: diverse disciplines including non-STEM, at least one Learning Center specialist, and a mix of local Colombian and international faculty — ensuring the fellowship tools are tested across disciplinary and cultural contexts from the outset.

Department Leadership is a Year 2 goal, not a Year 1 deliverable

Year 1 tests and refines the tools across diverse teacher perspectives and generates documented evidence of what works. The AI Task Force then brings the proven framework to department leadership in Year 2 as a ready, evidence-based structure rather than a prototype.

7.3 Two-Cycle Structure

The fellowship is organised around two iteration cycles, each reducing Cajina's structural role as teacher capacity grows:

Cycle	Structure and Cajina's Role
Cycle 1 — Redesign with Support (Aug–Feb)	Teacher selects one existing assessment and completes an initial VISTA audit alone. Cajina provides written feedback before enactment on AI-related design decisions. Content and pedagogy remain the teacher's domain. Teacher and Cajina co-plan: Cajina contributes AI-TPACK knowledge; teacher leads on content and students. Where appropriate, Cajina co-teaches or co-observes. After enactment, teacher reviews student reasoning artifacts. VSR Session 1: teacher and Cajina examine a classroom clip. Central question: What did I think would happen here, and what actually happened? Revision follows.
Cycle 2 — Supported Independence (Feb–Jun)	Teacher redesigns and plans independently — no co-planning, no co-teaching. Before enactment, Cajina provides one structured written feedback pass on the redesign: specifically on AI-related design decisions and visible reasoning strategy (content and pedagogy remain the teacher's domain). Teacher enacts independently. VSR Session 2: peer-facilitated by the pair partner — Cajina attends as a silent moderator, intervening only if a significant gap is being missed. No co-facilitation. Student artifacts reviewed. In June, each teacher brings their two-cycle arc to their own department.

Why Two Cycles Matter

Cycle 1 tests whether the tools work with support. Cycle 2 tests whether the learning can be transferred to independent practice — with structured written feedback before enactment and Cajina present as a moderator (not a leader) in VSR Session 2. The gap between Cycle 1 and Cycle 2 practice is the fellowship's most important measurement point.

7.4 Video-Stimulated Recall — The Central Reflection Mechanism

VSR is the fellowship's primary mechanism for examining the gap between what a teacher intended and what actually happened in the classroom — the most productive site for professional learning in the design. It is used once per cycle — twice per year — after every assessment redesign, not as a one-time event.

Intended vs. Enacted TPACK: When a teacher designs a task, they carry assumptions about how students will engage with AI. When they enact it, something different happens. VSR asks: what did I think would happen here? What actually happened? What does the gap reveal about how AI actually behaved in this discipline and task? Tiernan (2016) establishes that this gap is where the most significant professional learning occurs. The fellowship uses VSR in both cycles: Cajina-facilitated in Cycle 1, peer-facilitated in Cycle 2.

The department transfer session is where individual AI-TPACK becomes **collective knowledge** — or fails to. Documenting what happens when a teacher brings sustained disciplinary work back to colleagues who received none of the scaffolding is the bridge to Year 2 scaling.

7.5 The Augmented Human Thinking Routine — Digitising the Coaching Conversation

The Augmented Human Thinking Routine is the second tab of the VISTA website (vista-fellowship.vercel.app). It is both a coaching tool Cajina uses with teachers during Cycle 1 redesign sessions, and an independent thinking routine teachers run alone from Cycle 2 onward.

How Cajina uses it in Cycle 1. During co-redesign, Cajina walks the teacher through the Augmented Human Thinking Routine using AI-TPACK as the coaching lens. The teacher puts their own task into an AI tool the way a student would. They then work through four structured moves together: What can AI do with this task? How much of the output can you trust? What would happen in your classroom with your students? Who does what — student or AI? Each move starts with the teacher's own thinking before any AI is opened. Cajina's role is to surface the AI-TPACK dimension that is most relevant to that teacher's specific discipline and task: what does AI do to the cognitive demand, the SAIL positioning, the assessment validity? That conversation is what the Augmented Human Thinking Routine makes structured and repeatable.

How teachers use it independently in Cycle 2. By Cycle 2, teachers run the Augmented Human Thinking Routine on their own, without Cajina. The digital tool replaces the in-person coaching conversation. The teacher still thinks first — answering each of the four moves before opening any AI tool — then uses AI to extend and challenge what they saw. The routine builds the habit: not AI dependence, but the critical, independent judgment that uses AI as a thinking partner. The test of whether the fellowship worked is whether teachers run this routine alone, on any task, without prompting.

What it builds. The Augmented Human Thinking Routine develops AI-TPACK experientially rather than abstractly. Teachers learn what AI does to their specific assessments in their specific subjects by doing it, not by reading about it. Each time they run the routine, they add a data point to their own understanding of where AI changes the assessment equation in their discipline. By the department transfer sessions in June, teachers have a concrete, documented record of that learning.

7.6 The Fellowship as a Learning Arc — Key Moments and Milestones

Every scheduled touchpoint, format, duration, and output:

Month	Event	Format	Duration	Role & Output
August	Launch + Assessment Selection	Full cohort	60 min	Cajina introduces the fellowship, models a VISTA audit live on one of his own assessments, and demonstrates the Augmented Human Thinking Routine. Each teacher identifies the assessment they will redesign in Cycle 1. Output: assessment selected, initial VISTA audit begun.
September	PLC 1 — Initial Audit Calibration	Full cohort	60 min lunch	Provided lunch. Teachers share their initial VISTA audits. Cross-disciplinary discussion of what counts as interpretable evidence of learning. Redesign priorities surfaced and refined. Output: audit calibrated, redesign direction agreed.
October	Cycle 1 Enactment	Classroom	Class time	Teacher enacts the redesigned assessment. Student artifacts collected: reasoning traces, revision notes, AI decision logs, and any multimodal reflections (audio, video, embodied, creative) appropriate to the discipline.

Month	Event	Format	Duration	Role & Output
October (late)	PLC 2 — Early Artifact Analysis	Full cohort	60 min lunch	Provided lunch. Pairs share early student artifacts. Emerging patterns surfaced. Cross-disciplinary calibration: what evidence of learning are we actually seeing? Revision planning. Output: emerging patterns documented.
November	VSR Session 1	Individual + Cajina	30 min each	Teacher and Cajina review a short classroom clip together. Focus: intended versus enacted design. Where did the redesign produce the evidence the teacher hoped for? Where did it not? Gap analysis. Output: one concrete revision decision for the second half of Cycle 1 or for Cycle 2.
December	PLC 3 — Cycle 1 Synthesis	Full cohort	60 min lunch	Provided lunch. Cycle 1 synthesis across pairs. Cross-disciplinary tensions named. Redesign insights shared. Cycle 2 preparation begins. Output: Cycle 2 assessment direction agreed.
January	Cycle 2 Independent Redesign	Individual	~1 hr	Teacher completes the VISTA audit independently on the second assessment. Runs the Augmented Human Thinking Routine alone, without Cajina. One written feedback pass from Cajina on AI-related design decisions only — content and pedagogy remain the teacher's domain.
February	PLC 4 — Design Calibration	Full cohort	60 min lunch	Provided lunch. Cycle 2 design calibration through peer critique. AI-TPACK discussion grounded in concrete design decisions. Preparation for independent enactment. Output: Cycle 2 designs critiqued and refined.
February–March	Cycle 2 Enactment	Classroom	Class time	Teacher enacts independently. No co-teaching. Cajina available on request but does not initiate. Student artifacts collected.
March	VSR Session 2	Peer-facilitated	30 min	Peer-facilitated within the pair. Cajina attends as a silent moderator. Focus: independent enactment reflection and growth from Cycle 1 to Cycle 2.

Month	Event	Format	Duration	Role & Output
April	PLC 5 — Cross-Cycle Analysis	Full cohort	60 min lunch	Provided lunch. Cross-cycle comparison of evidence. Identify transferable practices. Begin department transfer planning. Output: transferable practices documented per teacher.
May	PLC 6 — Department Transfer Prep	Full cohort	60 min lunch	Provided lunch. Department transfer preparation. Departmental resources finalised. Showcase preparation. Output: transfer plan and showcase materials ready.
June	Showcase + AI Task Force Handoff	Full cohort + CNG community	~90 min	Each teacher presents their two-cycle arc, transferable practices, and student artifacts. Cajina presents cross-cohort findings. AI Task Force receives the complete handoff package, Knowledge Hub, and CNG chatbot specification. Output: public record; sustainability documented; handoff complete.

7.7 Communication Check-In Schedule

Beyond scheduled touchpoints, the fellowship includes structured **async check-in moments** at specific points in the year. These are brief — a short message or voice note, not a meeting — and serve two purposes: checking workload and wellbeing, and catching confusion or resistance before it compounds. They reflect Principle 11 from Whitman and Kelleher (2016): chronic stress impairs learning, and this fellowship is designed for real teachers in real schools.

Check-In Moment	Purpose and Method
After Enactment 1 (late September)	Cajina sends brief async message: How did it go? What surprised you? What would you do differently? No formal response required — teachers may reply with a voice note, a few lines, or simply note it in their reflection. Purpose: catch early friction before Redesign 2.
After Redesign 2 (mid-November)	Cajina checks in on workload: Is the pace sustainable? Are the feedback cycles useful? Any support needed before PLC Cycle 2? Purpose: explicit permission to flag problems rather than persist silently.

After VSR Cycle (late February)	Cajina checks in on the VSR experience: Was reviewing your own video useful or uncomfortable? What did you notice about yourself as a teacher? Purpose: VSR can surface unexpected feelings about one’s own practice — this check-in normalizes that.
Before Knowledge Hub Contribution (early April)	Cajina checks in on exemplar selection: Are you confident choosing your best work? Any exemplars you’re unsure about? Purpose: teachers with the most growth often undervalue their best artifacts.
Before Showcase (late May)	Cajina checks in on presentation readiness: What’s the one thing you most want the CNG community to understand from your year? Purpose: narrative preparation, not slide preparation — grounded in what the teacher actually learned.

7.8 Decision Rules for Visible Reasoning

Visible reasoning is strategic, not universal. *No AI and Full AI are not moral opposites.* The question is whether the task design still captures the intended learning.

Condition	Guidance
Use VR when...	AI can bypass threshold concept learning; reasoning quality matters more than product quality; misconceptions are likely hidden in polished outputs; transfer is a key outcome
Do not require VR when...	Task focuses on routine fluency; reflection adds workload without learning benefit; thinking is already visible through another evidence form; students would experience reflection fatigue

7.9 The Right to Say No — Non-AI Pathways, Student Consent, and Agency

This fellowship takes a firm position: **no student should be required to use AI as a condition of participating in learning.** Students may decline AI use for reasons of personal conviction, family values, religious belief, data privacy concern, or simply preference. These are not exceptions to manage — they are legitimate expressions of student agency. Respecting a student’s right to say no to AI is not a compromise of this fellowship’s goals. **It is one of its clearest expressions.**

The VISTA audit process makes this practical rather than theoretical. Every assessment redesigned through the fellowship must pass one test: a student who completes the task without AI must be doing the same intellectual work as a student who uses AI deliberately and documents their reasoning. If the non-AI version is significantly harder, the AI-assisted version has a design problem — not a student compliance problem. The task without AI is not the penalty version. It is the original.

The VISTA Paper Audit is the fellowship’s concrete instrument for this commitment. It is a printable, bilingual, one-page version of the same six-step VISTA audit — identical questions, identical rating scales, identical SAIL positioning, no browser or AI tool required. A teacher working with a student who declines AI use runs the same audit, makes the same redesign decisions, and collects the same category of reasoning evidence. The research design is not compromised. The student is not disadvantaged. The pathway is genuinely equivalent.

This distinction also has implications for how AI literacy is taught. Students can develop critical understanding of AI — examining outputs, evaluating AI-generated work, discussing the nature and limits of generative systems — without ever being required to produce AI-assisted work themselves. The difference between *understanding AI critically* and *using AI to complete tasks* is one this fellowship preserves deliberately. It is the difference between education about the world and a mandate to participate in it on terms students did not choose.

Design Test

If a student says ‘I don’t want to use AI for this’ and the task falls apart — the task was not designed well enough. A well-designed task in this fellowship is one where the answer to that student is: ‘Good. Here is what I need you to show me.’

7.10 Multiple Ways of Knowing

The fellowship applies Universal Design for Learning (CAST, 2018) as its accessibility framework. Interpretable evidence is accepted in multiple modes: verbal, written, visual, computational, **embodied, artistic, and creative**. Interaction traces — revision histories, making logs, prompt sequences — are explicitly valued alongside verbal explanation. The quality of reasoning is assessed, not the mode of expression.

7.11 Teacher-Friendly Language

The theoretical constructs in this proposal are the academic framing for design and research. In practice, during PLC sessions and everyday teacher conversation, accessible language is used:

Academic Term	Teacher-Facing Language
AI-TPACK	Does AI handle my subject's knowledge structures the way I do?
Ethical TPACK	Should I use AI here — and what could go wrong?
Contextual Knowledge (XK)	Will this work with my students, at CNG, right now?
HAIC-K	How do I set up the task so students stay in charge of the thinking?
VISTA Protocol	If AI did this perfectly, what learning would disappear?
Interpretable Evidence	What would I see that tells me the student actually understood?
Visible Reasoning	How can I make thinking visible — not just the answer?

7.12 Why AI? Why an AI Fellowship for an Assessment Problem?

If this is fundamentally an assessment validity problem, why is it an AI fellowship? Because AI is what made the problem unignorable, and because designing assessments that hold up in AI-rich classrooms requires specific knowledge that general assessment training does not provide. AI did not create the problem of polished products masking weak learning.

Historically, polished products often functioned as proxies for understanding — sometimes accurate, often not. AI made those proxies unreliable across almost every task we used to trust. AI created the urgency. The underlying challenge — generating interpretable evidence of learning — is older. Teachers in AI-rich classrooms need both: the broader assessment design capacity, and the specific knowledge of what AI does to that capacity in their discipline. This fellowship develops both.

7.13 The Visibility Claim: What This Fellowship Does and Does Not Claim

The fellowship's name — Visibility Into Student Thinking and Agency — can be misread as claiming that all thinking can or should be made visible. It does not. Tacit knowledge, artistic judgment, embodied knowing, mathematical intuition, and expert automaticity often resist explicit articulation. Teachers know this. The goal of VISTA is not complete visibility. It is **sufficient evidence to make stronger inferences about learning than product-only assessment allows**. A redesigned assessment does not need to make all of a student's thinking observable. It needs to generate enough interpretable evidence that the teacher can make a defensible judgment about understanding — even when AI may have been involved in producing the final product.

7.14 Failure Conditions: Evidence That Would Challenge the Fellowship's Assumptions

This fellowship operates as practitioner inquiry. That requires naming, in advance, the evidence that would suggest the design is not working. The following conditions would constitute meaningful disconfirmation:

- Teacher confidence in their assessment redesigns increases, but the quality of evidence collected does not.
- Student artifacts from redesigned assessments reveal little more about learning than artifacts from the original versions.
- Teachers abandon VISTA, the Augmented Human Thinking Routine, or PLC structures after the fellowship year ends.
- Departments fail to adopt the shared language or design practices teachers bring back from transfer sessions.
- Evidence collected through the redesigns remains difficult to interpret — either because the redesign was too ambitious or because the evidence captures noise rather than signal.

These are not hypothetical risks to manage rhetorically. The PLC and VSR structures explicitly examine the artifacts against these failure conditions. If they appear, the fellowship's findings will document them as honestly as it documents its successes.

7.15 Tool-Agnostic by Design

This fellowship is intentionally tool-agnostic. Its value does not depend on ChatGPT, Gemini, Claude, Flint, NotebookLM, or any AI platform that exists today or emerges next year. The central question — how to assess reasoning, interpretation, revision, and judgment when AI can produce sophisticated products — remains relevant regardless of which AI tools students use. VISTA, the Augmented Human Thinking Routine, and the assessment-redesign protocols are tool-independent. They will need updating as AI capabilities change, but the framework does not. That is part of why this fellowship is institutional infrastructure rather than a temporary response to current tools.

7.16 Addressing Resistance: Disciplined Inquiry, Not Ideological Agreement

Teachers entering this fellowship will hold different beliefs about AI, about assessment, about visible reasoning, and about student agency. That is expected and welcomed. The fellowship is structured as disciplined inquiry into assessment design questions — not as a requirement to adopt predetermined conclusions about AI. A teacher can be deeply skeptical of AI and still complete VISTA audits productively. A teacher who is enthusiastic about AI integration will still encounter the assessment validity problem in their own work. The fellowship’s tools function whether the participant arrives convinced or doubtful. What participants must commit to is the inquiry: examining their own assessments, collecting evidence, reflecting honestly, and revising in response to what they see. The conclusions are theirs to draw.

8. CNG Tool Ecosystem

Guiding Principle

Tools should be selected according to the thinking being developed. Not vice versa.

Three institutionally supported tools serve distinct pedagogical roles — described here as functions rather than platforms, because the function is what matters:

Tool	Pedagogical Role	Use When	Risks: Fluency, Bias, Privacy
Gemini Education (ChatGPT / Claude as alternatives)	<i>Divergent Thinking Partner Brainstorming, perspective-taking, critique generation</i>	Generating alternatives; exploring perspectives; producing material for student critique	Fluency illusion; premature closure; bias reproduction (training data reflects dominant perspectives); privacy (no PII in prompts)
NotebookLM	<i>Knowledge Hub & Evidence Grounding Partner Source triangulation, institutional memory</i>	Grounding claims in specified sources; source triangulation; evidence synthesis; SRP research preparation; PLC knowledge archiving	Source overreliance; false confidence; equity gap (source diversity shapes analysis quality)
Flint (teacher-discretion tool)	<i>Structured Inquiry Partner Guided questioning, visible reasoning</i>	Structured inquiry and Socratic coaching; teacher-designed learning experiences; assessment contexts with teacher-set parameters	Over-scaffolding; compliance orientation; design bias (teacher parameters shape what thinking is valued)

8.1 Technology Position Statement

This fellowship is intentionally **tool-agnostic**. Assessment routines are designed to remain valid regardless of future technological changes. When better tools become available, the HCAP question provides the evaluation criterion: does this serve human learning, or does it substitute for it?

Embedded ethics is not a separate step. Every tool integration is evaluated for bias in outputs, data privacy implications, and equitable design — embedded in the Assessment Audit, the Disclosure Protocol, and every PLC.

8.2 Backwards Design as the Iteration Logic

Every iteration in the fellowship is structured around backwards design (Wiggins & McTighe, 1998, 2005): teachers begin by identifying the specific disciplinary capacity the task is designed to develop, define what interpretable evidence of that capacity would look like in student work, and only then examine whether the current task design actually requires students to demonstrate it. The VISTA audit operationalizes this sequence explicitly: Step 1 asks teachers to name the learning and describe the evidence *before* asking anything about AI.

Backwards design reveals a structural problem in AI-rich environments: many tasks were originally designed for a pre-AI context and were often not rigorously evidence-first to begin with. AI does not create the problem of tasks that don't require the thinking they claim to require — it makes the problem impossible to ignore. The fellowship uses that visibility as a pedagogical opportunity. Every redesign is also a backwards design exercise. Every teacher who completes an iteration cycle is developing the professional habit of starting from the question: *What would I see in student work that tells me this learning actually happened?*

9. Cognitive Agency Knowledge Hub

The primary long-term knowledge infrastructure is the **Cognitive Agency Knowledge Hub** — a shared NotebookLM workspace that evolves continuously. A document becomes outdated. A knowledge hub grows with each iteration, each new exemplar, each PLC discussion archived.

The Hub contains:

- Research library: foundational and current literature on AI-TPACK, visible reasoning, assessment design, and PLC sustainability
- Annotated exemplars: student reasoning artifacts organized by discipline and evidence mode
- Assessment audits: documented VISTA Protocol applications across disciplines
- Visible reasoning protocols: nine protocols with purpose, use cases, and sample prompts
- AI reflection and disclosure examples: authentic, avoids performative compliance
- VSR examples: hinge point analysis with teacher commentary
- PLC discussion archives: epistemological tension discussions and cross-disciplinary patterns
- Implementation resources: AI Task Force Handoff Package, Department PLC Protocol, Curriculum Review Integration guide

Why NotebookLM specifically: NotebookLM enables source-grounded inquiry — teachers and Task Force members can ask questions of the Hub’s content and receive evidence-grounded responses. This models the same epistemic practices the fellowship promotes in students: ground your claims, triangulate across sources, interrogate outputs critically. The Hub does not replace teacher judgment. It supports it.

The Long-Term Trajectory: From Hub to Institutional Chatbot

The Cognitive Agency Knowledge Hub is designed with a long-term trajectory in mind. In its early form it is a shared NotebookLM workspace. Over time, as it accumulates research, exemplars, audits, PLC archives, and implementation experience, it develops into something more powerful: a **CNG institutional AI chatbot** — a queryable knowledge base that any teacher, department chair, or administrator can interrogate to get evidence-grounded guidance on assessment design in AI-rich classrooms.

The trajectory:

Stage	Hub Capacity
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Year 1	Shared NotebookLM workspace. Five teachers contributing. Research library and first exemplars. AI Task Force members as co-stewards.
Year 2	Growing repository across divisions. Department PLC archives added. Cross-disciplinary patterns documented. Task Force queries the Hub to prepare facilitation.
Year 3	LC, Makerspace, and support faculty contributions. Multimodal and UDL-adapted exemplars. Hub begins to serve as onboarding resource for new teachers.
Year 4	Institutional integration. The Hub is embedded in departmental curriculum review. Any teacher at CNG can query it: What does the evidence say about visible reasoning in mathematics? What exemplars exist for science CER tasks? What does the research say about AI and embodied learning?
Beyond Year 4	The Hub evolves toward a fully queryable CNG institutional chatbot — trained on CNG’s own professional learning, assessment research, exemplars, and disciplinary science of teaching literature. It becomes a living institutional memory that grows smarter with every cohort, every PLC discussion, every redesign decision documented.

This trajectory is not a technology ambition. It is a **learning organization ambition** — the vision of a school that captures its own professional wisdom, makes it queryable, and uses it to continuously improve. The fellowship is the first step in building that infrastructure, starting with the specific domain where CNG needs it most right now: assessment design in AI-rich classrooms.

10. Departmental PLC and Curriculum Integration

10.1 Why Departments Are the Right Unit

Sustainable educational change does not happen through individuals. Fullan (2016) demonstrates that deep change requires embedding new practices in collaborative professional structures. DuFour and Eaker (1998) establish that PLCs are the most powerful professional development model available. Coburn (2003) argues that sustainable reform requires *depth* — shifting underlying beliefs and practices. Bryk et al. (2015) demonstrate that reliable change builds on existing organizational structures. Hargreaves and Fullan (2012) identify *professional capital* — collective knowledge, skill, and decisiveness — as the key variable in educational improvement.

Departments already share **CK, PK, and PCK** — the foundational knowledge that makes AI-specific layers meaningful. A mathematics department that shares a commitment to proof and justification already knows what it means for AI to mishandle a proof. A humanities department that shares a commitment to evidence and interpretation already knows what a good historical argument looks like. The fellowship builds the AI-specific knowledge layers — AI-TPACK, Ethical TPACK, XK, and HAIC-K — on top of this shared foundation. This is not about replacing what teachers know. It is about extending it.

10.2 Department Cognitive Agency PLC Protocol

The most scalable output is a **repeatable 30-minute routine** any department can use independently:

1. Select one recent or upcoming assessment.
2. Ask: **If AI can produce the answer, how do we assess the reasoning, interpretation, revision, and judgment behind it?**
3. Identify the thinking most at risk of becoming invisible with AI assistance.
4. Identify the evidence that would demonstrate that thinking is genuine.
5. Identify the appropriate SAIL positioning for this task.
6. Make one specific change to the assessment design.

The protocol requires no new curriculum, no new technology, and no external facilitator once established. Its power comes from repeated use — each iteration building shared departmental language, shared exemplars, and shared judgment. Research identifies three mechanisms that make this routine durable: artifact analysis (teachers reviewing student work together develops shared disciplinary standards faster than any training), collaborative co-design (teachers who design assessments together own them and use them consistently), and collective sensemaking (teachers negotiating together what AI means for their discipline rather than resolving it individually). Sustainable change comes from these routines embedded in existing structures, not from isolated professional development events.

10.3 Curriculum Review Integration

The most durable innovations become embedded in existing systems. The fellowship’s assessment audit question can be incorporated into curriculum review through a single addition:

Curriculum Review Integration Question

What evidence will demonstrate learning if AI can generate the product? This moves the work from an AI initiative to a curriculum and assessment initiative — the reframe needed for long-term sustainability.

This question requires no new training and no new meeting. By Year 4, it is embedded in CNG’s departmental planning as standard practice.

10.4 Emerging Shared Language — Co-Constructed Through Inquiry

A key outcome of the departmental PLC work is the gradual co-construction of shared professional language — ways of talking about assessment, evidence, and student reasoning that travel across disciplinary boundaries. This language is not pre-designed and is not presented as a framework. It emerges from sustained collaborative inquiry. The June transfer sessions are one site where this language is observed: what terms prove transportable? What concepts require translation? That data becomes part of the fellowship’s contribution to the CNG professional community — the beginning of a vocabulary, not the announcement of one.

11. Deliverables

Full documentation for each deliverable appears in the **Appendices**. The following is a summary.

11.1 Per Teacher

- Two to three documented assessment iteration arcs
- Written feedback correspondence documenting I-CK development
- One VSR cycle with hinge point identification
- AI Reflection and Disclosure Protocol enacted with students
- Contributions to Cognitive Agency Knowledge Hub: two to four exemplars with documentation
- June showcase presentation: 10 minutes

11.2 Fellowship Cohort Products

- Assessment Audit Toolkit: VISTA Protocol (digital and paper), facilitation guide, discipline-specific examples (**Appendix A**)
- VISTA Paper Audit: printable, bilingual one-page audit instrument — identical to the digital tool, no browser or AI required. Ensures every teacher and student has a non-AI pathway generating equivalent evidence.
- Visible Reasoning Protocol Suite: nine protocols (**Appendix B**)
- Cross-Disciplinary Rubric: five dimensions, four levels, developmental progressions (**Appendix C**)
- Triangulated Evidence Framework: validity argument structure (**Appendix D**)
- AI Reflection and Disclosure Record (**Appendix E**)
- Department Cognitive Agency PLC Protocol (**Appendix F**)

12. Evaluation and Research Design

12.1 Primary Research Question

When teachers develop the capacity to identify and assess evidence of reasoning, interpretation, revision, and judgment in AI-rich classrooms, does that capacity change how they design assessment, what evidence they collect, and how they interpret what they see? The primary claim: teachers who complete two VISTA audit cycles develop more specific and defensible criteria for what counts as evidence of learning when AI is available.

12.2 Research Design Matrix

Mixed-methods practitioner inquiry. Data sources triangulate teacher capacity (TIAI measures, VISTA audit quality, VSR transcripts), student evidence (artifacts, reflections, interviews), and institutional capacity (PLC artifacts, transfer session evidence, departmental language adoption).

12.3 Validity Strategy — Triangulated Evidence

No single category of evidence settles the question. Validity emerges from the convergence of teacher self-report (TIAI), teacher artifact evidence (audits, redesigns, VSR transcripts), student evidence (multimodal artifacts and reflections), and institutional evidence (cross-cohort PLC patterns and transfer outcomes). Where these triangulate, claims are stronger. Where they diverge, the divergence itself becomes data.

12.4 Evidence Architecture: Teacher, Student, and Institutional

The fellowship's evidence model is deliberately multi-source. Three categories are collected and triangulated:

Teacher Evidence

- VISTA audit documents — analysis of each assessment before redesign
- Redesign artifacts — the assessment as taught, with explicit notes on what changed and why
- VSR reflections — transcribed analysis of intended versus enacted design
- Implementation notes — what actually happened during enactment

Student Evidence

- Reasoning artifacts — work that captures decision-making, not just outputs
- Revision narratives — written or spoken accounts of how thinking changed
- AI decision logs — student documentation of where, how, and why AI was used
- Interviews and surveys — structured student perspectives on the redesigned tasks
- Audio, video, and multimodal reflections
- Artistic, embodied, and creative responses — evidence that resists text-only capture

Institutional Evidence

- PLC artifacts — documented insights, tensions, and shared language emerging across pairs
- Transfer session records — language, questions, and resistance surfaced in department sessions
- Departmental language adoption — documented uptake of fellowship terminology and practices
- Collective AI-TPACK development — cross-cohort patterns

Multimodal student evidence is grounded in Universal Design for Learning (CAST, 2018): students demonstrate reasoning through multiple means of expression. Equitable evidence collection is a design requirement.

12.5 Scope of Year 1

Year 1 is intentionally bounded. Six teachers, two cycles, one academic year. The scope is to test the tools (VISTA, the Augmented Human Thinking Routine, the PLC structure) in real classrooms, generate documented evidence of what works and what does not, and prepare the Year 2 institutional version. Year 1 does not claim to scale to the whole school. It claims to produce the infrastructure that makes scaling possible.

13. Sustainability and Institutional Rollout

13.1 The Long-Term Vision

This fellowship is institutional infrastructure, not a one-year intervention. Year 1 builds the evidence base, refines the protocols, and produces the materials the AI Task Force and department leadership need to scale the work in Year 2 and beyond.

13.2 Year 2 Architecture: From Individual to Collective TPACK

Each Year 1 teacher becomes the lead facilitator for VISTA work within their own department. The AI Task Force receives the complete handoff package: protocols, instruments, facilitation guides, and the Year 1 evidence base. Department leadership joins the facilitation team. Cajina shifts from primary facilitator to consultant. The CNG Chatbot specification — grounded in Year 1's student reasoning exemplars and disciplinary standards — enters development.

13.3 Scaling Model

The scaling model is departmental, not whole-school. Each Year 1 teacher seeds the work in their own department. Departments adopt and adapt the tools based on disciplinary context. The fellowship resists mandated uniformity — the goal is that each department develops its own evidence-informed practice with shared vocabulary across the school.

13.4 Change Management

Year 1: cohort builds tools and evidence. Year 2: departments adopt with Year 1 teachers as facilitators. Year 3: practices become institutional norms supported by the AI Task Force, the CNG Chatbot, and the Knowledge Hub. This timeline reflects the research on educational change (Fullan, 2007; Bryk et al., 2015): durable change in complex systems takes years, not months.

14. Direct Alignment with CNG School Priorities

This fellowship aligns directly with CNG’s mission, vision, Strategic Directions, and the Innovation Learning Initiative.

- **CNG Mission and Vision:** developing responsible global citizens who achieve their fullest potential requires assessment that reliably surfaces what they understand. This fellowship develops the capacity to design that assessment in AI-rich conditions.
- **Strategic Directions — Shareable, Scalable, Replicable:** VISTA, the Augmented Human Thinking Routine, the PLC protocol, and the Knowledge Hub are designed for cross-disciplinary and cross-divisional transfer.
- **Science of Learning:** Every design decision is grounded in Dehaene (2020), Whitman and Kelleher (2016), Messick (1989), Sadler (1989), Boud and Falchikov (2007), Hattie (2009), and Timperley (2011).
- **Responsible and Ethical AI Use:** SAIL framework embedded in every redesign. AI Reflection and Disclosure Record (Appendix E). Student right to decline AI use is protected by design.
- **Student-Centered and Innovative Learning:** Multimodal student evidence, UDL-grounded design, student voice as a formal evidence source.
- **Innovation Learning Initiative:** Year 1 generates documented evidence. Year 2 scales institutionally. The CNG Chatbot specification offers a concrete tutorial-chatbot trajectory grounded in CNG’s own pedagogy.

15. References

- Black, P., & Wiliam, D. (1998). Inside the black box: Raising standards through classroom assessment. *Phi Delta Kappan*, 80(2), 139–148.
- Boud, D., & Falchikov, N. (2007). *Rethinking assessment in higher education: Learning for the longer term*. Routledge.
- Brown, A. L. (1992). Design experiments: Theoretical and methodological challenges in creating complex interventions in classroom settings. *Journal of the Learning Sciences*, 2(2), 141–178.
- Bryk, A. S., Gomez, L. M., Grunow, A., & LeMahieu, P. G. (2015). *Learning to improve: How America's schools can get better at getting better*. Harvard Education Press.
- CAST. (2018). *Universal Design for Learning Guidelines version 2.2*. <http://udlguidelines.cast.org>
- Collins, A. (1992). Toward a design science of education. In E. Scanlon & T. O'Shea (Eds.), *New directions in educational technology* (pp. 15–22). Springer.
- Dawson, P. (2021). *Defending assessment security in a digital world: Preventing e-cheating and supporting academic integrity in higher education*. Routledge.
- Dehaene, S. (2020). *How we learn: Why brains learn better than any machine ... for now*. Viking.
- Fullan, M. (2007). *The new meaning of educational change* (4th ed.). Teachers College Press.
- Hattie, J. (2009). *Visible learning: A synthesis of over 800 meta-analyses relating to achievement*. Routledge.
- Hattie, J. (2012). *Visible learning for teachers: Maximizing impact on learning*. Routledge.
- Messick, S. (1989). Validity. In R. L. Linn (Ed.), *Educational measurement* (3rd ed., pp. 13–103). Macmillan.
- Sadler, D. R. (1989). Formative assessment and the design of instructional systems. *Instructional Science*, 18(2), 119–144.
- Timperley, H. (2011). *Realizing the power of professional learning*. Open University Press.
- Whitman, G., & Kelleher, I. (2016). *Neuroteach: Brain science and the future of education*. Rowman & Littlefield.

— *End of Proposal* —